Q30: Comprehension Check

DLS 2026 — Kukla

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Comprehension check. During training of a deep network you hit out-of-memory, but the *model parameters* comfortably fit in GPU memory. What is most likely consuming the memory, and name a technique that reduces it by doing extra computation.

Answer. The **activations**—the intermediate forward-pass tensors that reverse-mode autodiff must keep alive to compute gradients in the backward pass—are most likely consuming the memory. Activation memory scales with network depth and batch size and typically dominates parameter memory during training. **Gradient checkpointing (rematerialization)** reduces it: instead of storing all activations, you save only a subset and *recompute* the others during the backward pass, trading a second partial forward (extra compute) for a large reduction in peak memory.